

CODECOMPASS EVALUATION

# Official Bilingual Embedding Model Comparison

A controlled 60-question retrieval study of local nomic embeddings and Gemini embeddings through AvalAI

EVIDENCE SUPPORTS GEMINI AS A HIGH-QUALITY OPTIONAL EMBEDDING CONFIGURATION

| HYBRID TOP-3   | PERSIAN HYBRID TOP-3 | SEMANTIC MRR@10  |
|----------------|----------------------|------------------|
| 78.3% -> 95.0% | 76.7% -> 96.7%       | 0.5061 -> 0.8264 |

60 questions / 30 bilingual concepts / 3 pinned Python repositories  
Only the embedding provider and model changed.

# Contents

| Section | Scope                            |
|---------|----------------------------------|
| 1       | Objective and controlled design  |
| 2       | Dataset and index identity       |
| 3       | Global retrieval results         |
| 4       | Persian and English results      |
| 5       | Repository and category slices   |
| 6       | Case transitions and regressions |
| 7       | Interpretation and limitations   |
| 8       | Reproducibility appendix         |

# 1. Objective and controlled design

The experiment tests whether the embedding model explains part of CodeCompass retrieval quality, particularly for Persian questions. It reuses the frozen official local baseline and creates an isolated Gemini treatment index from the same canonical SQLite chunks.

| Component          | Control                                           |
|--------------------|---------------------------------------------------|
| Dataset            | 60 fixed questions; 30 English/Persian pairs      |
| Repositories       | Flask, itsdangerous, MarkupSafe at pinned commits |
| Canonical evidence | 1,871 unchanged chunks and stable chunk IDs       |
| Retrieval          | Lexical, semantic, hybrid; Top-10; unchanged RRF  |
| Changed variable   | nomic local (768d) -> Gemini (3072d)              |
| Generation         | No LLM calls                                      |

## Fail-closed control

The treatment run was rejected unless all vector counts matched and every lexical Top-10 ranking was identical to the frozen baseline. This prevents accidental retrieval or corpus changes from being attributed to the embedding model.

## 2. Dataset and index identity

The official benchmark is balanced by language but not by repository: Flask contributes 30 questions, itsdangerous 20, and MarkupSafe 10. Therefore both global micro results and repository slices are reported.

| Repository           | Commit     | Files | Chunks / vectors |
|----------------------|------------|-------|------------------|
| pallets/flask        | d318b68347 | 83    | 1611 / 1611      |
| pallets/itsdangerous | 672971d66a | 15    | 144 / 144        |
| pallets/markupsafe   | b2e4d9c768 | 12    | 116 / 116        |

### Index controls

All treatment indexes contain exactly one vector per canonical chunk. The local arm uses the frozen official vectors; no local embedding or retrieval call was repeated. Gemini document and query vectors were stored only under the isolated experiment runtime.

### 3. Global retrieval results

Gemini materially improved semantic retrieval and strengthened the hybrid system. Lexical equality is an experimental control, not an improvement claim.

| Method   | Metric             | Local  | Gemini | Delta  |
|----------|--------------------|--------|--------|--------|
| Lexical  | Top-1              | 43.3%  | 43.3%  | 0.0%   |
| Lexical  | Top-3              | 71.7%  | 71.7%  | 0.0%   |
| Lexical  | MRR@10             | 0.5809 | 0.5809 | 0.0000 |
| Lexical  | Evidence recall@10 | 81.7%  | 81.7%  | 0.0%   |
| Semantic | Top-1              | 35.0%  | 71.7%  | 36.7%  |
| Semantic | Top-3              | 65.0%  | 93.3%  | 28.3%  |
| Semantic | MRR@10             | 0.5061 | 0.8264 | 0.3203 |
| Semantic | Evidence recall@10 | 71.7%  | 95.0%  | 23.3%  |
| Hybrid   | Top-1              | 63.3%  | 71.7%  | 8.3%   |
| Hybrid   | Top-3              | 78.3%  | 95.0%  | 16.7%  |
| Hybrid   | MRR@10             | 0.7322 | 0.8306 | 0.0983 |
| Hybrid   | Evidence recall@10 | 86.7%  | 95.0%  | 8.3%   |

#### Primary finding

Hybrid Top-3 rose by 16.7 percentage points and Hybrid MRR@10 rose by 0.0983. Semantic Top-1 rose by 36.7 percentage points, showing that the embedding replacement primarily improved semantic candidate discovery.

## 4. Persian and English results

Both languages improved, with a substantially larger gain for Persian. This directly addresses the project's bilingual retrieval objective.

| Lang. | Method   | Local Top-1 | Gemini Top-1 | Local Top-3 | Gemini Top-3 | Local MRR | Gemini MRR |
|-------|----------|-------------|--------------|-------------|--------------|-----------|------------|
| EN    | Semantic | 50.0%       | 73.3%        | 76.7%       | 93.3%        | 0.6356    | 0.8444     |
| EN    | Hybrid   | 70.0%       | 73.3%        | 80.0%       | 93.3%        | 0.7867    | 0.8389     |
| FA    | Semantic | 20.0%       | 70.0%        | 53.3%       | 93.3%        | 0.3767    | 0.8083     |
| FA    | Hybrid   | 56.7%       | 70.0%        | 76.7%       | 96.7%        | 0.6778    | 0.8222     |

### Persian result

Persian Semantic Top-1 increased from 20.0% to 70.0%, Top-3 from 53.3% to 93.3%, and MRR@10 from 0.3767 to 0.8083. Persian Hybrid Top-3 reached 96.7%. The result indicates that the local embedding model was a material Persian alignment bottleneck on this dataset.

## 5. Repository and category slices

The improvement appears across all three repositories and most categories. The category analysis also preserves the main negative finding.

| Slice             | Method   | Local Top-3 | Gemini Top-3 | Local MRR | Gemini MRR |
|-------------------|----------|-------------|--------------|-----------|------------|
| direct_symbol     | Hybrid   | 75.0%       | 100.0%       | 0.5069    | 0.8056     |
| direct_symbol     | Semantic | 50.0%       | 91.7%        | 0.3833    | 0.8542     |
| function_behavior | Hybrid   | 79.2%       | 95.8%        | 0.8174    | 0.8681     |
| function_behavior | Semantic | 66.7%       | 95.8%        | 0.5542    | 0.8958     |
| multi_symbol      | Hybrid   | 75.0%       | 91.7%        | 0.7250    | 0.7153     |
| multi_symbol      | Semantic | 83.3%       | 83.3%        | 0.5306    | 0.6667     |
| semantic_behavior | Hybrid   | 83.3%       | 91.7%        | 0.7944    | 0.8958     |
| semantic_behavior | Semantic | 58.3%       | 100.0%       | 0.5083    | 0.8194     |
| flask             | Hybrid   | 80.0%       | 100.0%       | 0.7594    | 0.8722     |
| flask             | Semantic | 56.7%       | 96.7%        | 0.4856    | 0.8694     |
| itsdangerous      | Hybrid   | 75.0%       | 90.0%        | 0.7200    | 0.7458     |
| itsdangerous      | Semantic | 80.0%       | 85.0%        | 0.5633    | 0.7000     |
| markupsafe        | Hybrid   | 80.0%       | 90.0%        | 0.6750    | 0.8750     |
| markupsafe        | Semantic | 60.0%       | 100.0%       | 0.4533    | 0.9500     |

### Mixed multi-symbol result

For multi-symbol questions, Hybrid Top-3 improved from 75.0% to 91.7% and evidence recall improved, but Top-1 fell from 58.3% to 50.0% and MRR@10 fell from 0.7250 to 0.7153. Gemini is therefore not uniformly better at ranking every multi-evidence target.

## 6. Case transitions and regressions

| Method   | Recovered | Improved | Stable | Lower | Lost |
|----------|-----------|----------|--------|-------|------|
| Lexical  | 0         | 0        | 60     | 0     | 0    |
| Semantic | 13        | 18       | 20     | 9     | 0    |
| Hybrid   | 4         | 14       | 36     | 6     | 0    |

### Recovered Hybrid targets

| Recovered Hybrid case             | Lang. | Repository   | Gemini rank |
|-----------------------------------|-------|--------------|-------------|
| flask_after_request_processing_fa | FA    | flask        | 2           |
| flask_ensure_sync_fa              | FA    | flask        | 3           |
| itsdangerous_sign_value_en        | EN    | itsdangerous | 3           |
| itsdangerous_sign_value_fa        | FA    | itsdangerous | 3           |

### Hybrid rank regressions

| Hybrid case                        | Lang. | Category          | Local | Gemini |
|------------------------------------|-------|-------------------|-------|--------|
| flask_method_view_dispatch_fa      | FA    | function_behavior | 1     | 2      |
| flask_route_registration_fa        | FA    | multi_symbol      | 1     | 2      |
| flask_session_cookie_round_trip_en | EN    | multi_symbol      | 2     | 3      |
| itsdangerous_fallback_unsigners_fa | FA    | multi_symbol      | 1     | 2      |
| itsdangerous_verify_signature_en   | EN    | function_behavior | 1     | 2      |
| markupsafe_unescape_entities_en    | EN    | function_behavior | 1     | 2      |

All six regressions were one-rank movements and remained within Top-3. No Hybrid or Semantic target that was previously found in Top-10 was lost from Top-10.



## 7. Interpretation and limitations

The larger official benchmark confirms the earlier 18-case direction: embedding capability was an important retrieval constraint, especially for Persian. Gemini recovered 13 Semantic targets and 4 Hybrid targets that the local model missed within Top-10.

The evidence supports offering Gemini as an optional quality-oriented embedding configuration. It does not justify silently replacing the local default because cloud use changes privacy, availability, cost, and reproducibility properties.

### Limitations

| Limitation                | Consequence                                                                |
|---------------------------|----------------------------------------------------------------------------|
| Three Python repositories | Generalization to other languages and domains is unmeasured.               |
| Thirty concept pairs      | Results are descriptive; paired languages are not 60 independent concepts. |
| No downstream QA run      | No answer-quality improvement is claimed.                                  |
| Different runtime dates   | Provider latency is not compared.                                          |
| Multi-symbol mixed result | Top-rank precision remains imperfect.                                      |
| External API              | Availability, privacy, and cost differ from local execution.               |

## 8. Reproducibility appendix

| Artifact              | Canonical identity                                               |
|-----------------------|------------------------------------------------------------------|
| Benchmark             | 2a04a4f1b707481126c31673840670b4b72d3877c34b1990f12b2245688d69aa |
| Frozen local baseline | 45c0b3fb1adb91224e24cf8a9f42611e632afcfb5cf4d492518492ffbe700edc |
| Snapshot manifest     | 31ddf2c8c9de4649662ed9a4ada04c06d8473b733d23aa2ce29419301e3090b0 |

### Execution record

| Operation                  | Count |
|----------------------------|-------|
| Local embedding calls      | 0     |
| Gemini document embeddings | 1,871 |
| Gemini query embeddings    | 60    |
| Retrieval records per arm  | 180   |
| LLM calls                  | 0     |

### Artifact locations

Raw predictions: reports/evaluation/official\_embedding\_comparison\_v1/retrieval\_results.json  
Derived metrics: reports/evaluation/official\_embedding\_comparison\_v1/comparison\_summary.json  
Experiment identity: reports/evaluation/official\_embedding\_comparison\_v1/experiment\_manifest.json

### Final conclusion

On the frozen official bilingual benchmark, Gemini produced a meaningful retrieval improvement and a particularly large Persian semantic gain. The local embedding path remains the privacy-preserving offline option; Gemini is the stronger measured quality option for this evaluation scope.